

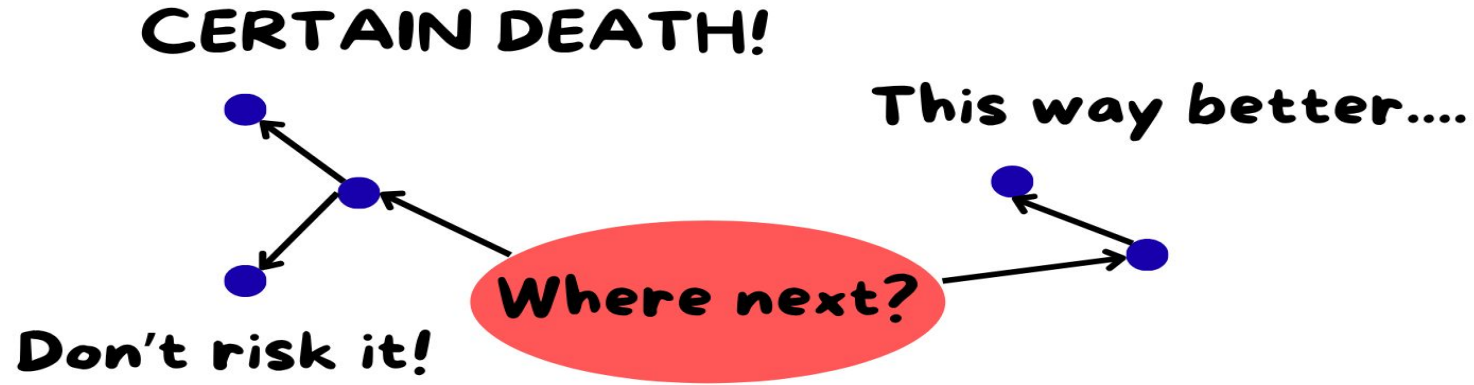
Quantum Beam Search (QBS)

Rød grød med fløde

Group 6

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How to choose where to go



Techniques

- Graph classical pruning
- 2- length path detection at every node
- Tensor Network inspired route reranker
- Quantum path chooser through amplitude amplification
- Scaling not so bad in terms of qubits (N possible paths require $\log N$ qubits), but it requires state preparation and a complex unitary implementation :(

FIGURE 1: Simplified Diagram of the Hybrid Quantum-Classical Hike Optimization Pipeline

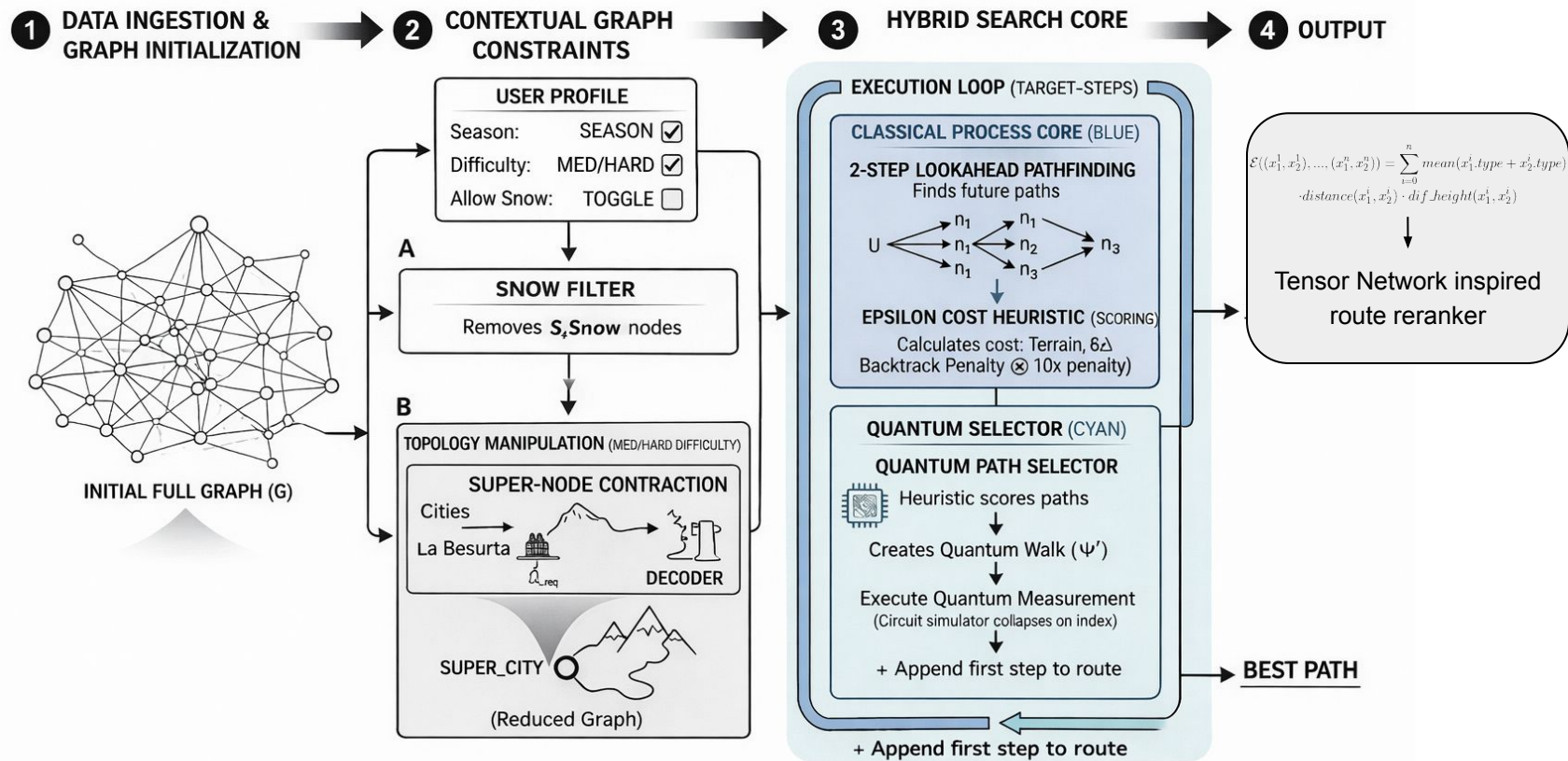


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Try it yourself!